UNIT 1 · TEACHER PLAYBOOK

Energy Transfer & Transformation

Grade 4 · CKSci-aligned · Teacher Edition

TEACHER PLAYBOOK

SCOPE Phase 5 — Chapter 1 supporting-resource ecosystem complete. Guided Notes + Slideshow are FINAL (Phases 3–4); practice/application, chapter quiz, absent-recovery, and FLEX Energy Detectives are retained; study-guide contribution + assessment trace are recorded. Chapters 2–7 spreads and special sections remain PLANNED placeholders pending later phases.

AUTHORITY Authoritative curriculum = CKSci Student Reader + Teacher Guide. This playbook documents the final system; it does not drive curriculum decisions.

UNIT AT A GLANCE

FIELD	DETAIL
Unit title	Energy Transfer & Transformation (Grade 4)
Chapters	1 Energy Causes Change · 2 Moving Objects Have Energy · 3 Energy and Speed Are Related · 4 Energy Transfer · 5 Collisions Transfer Energy · 6 How Can Energy Transformations Solve Problems? · 7 Solving Problems and Designing Solutions: Thomas A. Edison
Unit Big Idea	Energy can be transferred (moved) and transformed (changed in form). Both produce observable changes that are evidence of energy.
Transfer vs transformation	Transfer = energy moves (place/object). Transformation = energy changes form. BOTH is valid — never force either/or.
Major vocabulary	14 Core Vocabulary terms (see Unit Vocabulary Map): change, energy, stored energy, motion, energy of motion, speed, energy transfer, collide, collision, sound, convert, conversion, energy transformation, engineering design.
Flagship lab	Ramp It Up 2.0 (Ch3 · 2-day quantitative investigation)
Required build	Pom-Pom Launcher (end of unit · formative)
Assessment	Unit 1 final = 25 questions / 50 points (matching Q1–8 · multiple choice Q9–25) + separate teacher key.
Duration	Weeks 1–4 (16 core blocks, 30-min each) + Friday FLEX
Primary student resources	Guided Notes (primary) · Science Journal (secondary) · practice/application · labs · study guide · review game

DATED PACING CALENDAR — WEEKS 1–4

30-minute blocks · M–Th core + Friday FLEX. Slides/notes are PLANNED placeholders until Phases 3–4 land.

WEEK 1 · 10/12–10/16 · Chapters 1–2

DAY	CHAPTER / ACTIVITY	READER	GUIDED NOTES	DEMO / ACTIVITY	COLLECT / GRADE
Mon 10/12	Ch1 Block 1 — Observe & Name	1–2	Key Words · Model It · Draw/Label	Sharpener DEMO	exit ticket
Tue 10/13	Ch1 Block 2 — Forms & Stored	3–5	Forms · Stored · Evidence table	Flashlight + salt demos	exit ticket
Wed 10/14	Ch2 Block 1	7–9	Key Words · Model It · Draw/Label	Marble Collision DEMO	exit ticket
Thu 10/15	Ch2 Block 2	10–12	Before/After · Example/Non-example	—	exit ticket
Fri 10/16	FLEX — RECOVER/PRACTICE/EXTEND	—	—	Energy Detectives · Moving Marbles (FLEX)	not graded

WEEK 2 · 10/19–10/23 · Chapter 3 + Ramp It Up 2.0

DAY	CHAPTER / ACTIVITY	READER	GUIDED NOTES	DEMO / ACTIVITY	COLLECT / GRADE
Mon 10/19	Ch3 Block 1	13–14	Key Words · Model It	Ramp + Target Preview DEMO	exit ticket
Tue 10/20	Ramp It Up 2.0 —	13–14	Data table (lab)	LAB: calibration · 3	lab sheet

DAY	CHAPTER / ACTIVITY	READER	GUIDED NOTES	DEMO / ACTIVITY	COLLECT / GRADE
	Day 1			heights × 3 trials · cm	(formative)
Wed 10/21	Ramp It Up 2.0 – Day 2	13–14	Graph: Ramp Height vs Avg Cup Displacement	LAB: averages · graph · Design-Your-Ramp	lab + graph (formative)
Thu 10/22	Ch3 synthesis	15–16	Conclusion · Fair Test	evidence-based conclusion	exit ticket
Fri 10/23	FLEX	–	–	Drop It (FLEX) · graph/data practice	not graded

WEEK 3 · 10/26–10/30 · Chapters 4–5

DAY	CHAPTER / ACTIVITY	READER	GUIDED NOTES	DEMO / ACTIVITY	COLLECT / GRADE
Mon 10/26	Ch4 Block 1 – Sound + Light	17–22	Key Words · Source→Path→Receiver	Balloon Drum / salt DEMO	exit ticket
Tue 10/27	Ch4 Block 2 – Heat + Electricity	23–24	Source→Path→Receiver organizer	Source→Path→Receiver organizer (ACTIVITY)	organizer (formative)
Wed 10/28	Ch5 Block 1	25–27	Key Words · Before/After	Toy-Car Collision DEMO	exit ticket
Thu 10/29	Ch5 Block 2	28–30	Before/After diagram · Scenarios	collision scenarios (ACTIVITY)	application (formative)
Fri 10/30	FLEX	–	–	Newton's cradle demo (teacher) · collision predictor	not graded

WEEK 4 · 11/2–11/6 · Chapters 6–7 + Pom-Pom + review + assessment

DAY	CHAPTER / ACTIVITY	READER	GUIDED NOTES	DEMO / ACTIVITY	COLLECT / GRADE
Mon 11/2	Ch6 Block 1 — Transformations	31–33	Key Words · Input→Output model	Wind-Up Timer DEMO	exit ticket
Tue 11/3	Ch6 Block 2 — Engineering Design	34–36	Engineering cycle · criteria/constraints	T/T/BOTH sort (ACTIVITY)	exit ticket
Wed 11/4	Ch7 — Edison / Unit Synthesis	37–42	Unit Concept Map	Flashlight DEMO	exit ticket
Thu 11/5	Pom-Pom Launcher (REQUIRED BUILD) + review	—	—	build + test/improve · study guide	build (formative)
Fri 11/6	Review + Unit 1 assessment	—	—	review game · 25Q/50pt written final	GRADED (25 Q / 50 pts)

PACING FLAG Week 4 compresses Ch6 (2 blocks) + Ch7 (1 block) + Pom-Pom build + review + assessment into 5 days. Feasible only with the compact in-unit engineering cycle (RD-4) and a streamlined review. The 25Q/50pt final is ~30–45 min and may split Thu/Fri. See docs/playbook/UNIT1_PACING_CALENDAR.md.

HOW TO USE THIS UNIT

ELEMENT	WHAT TO KNOW
Blocks	30-minute science blocks · M–Th core + Friday FLEX. No essential new core content on Friday.
Primary artifact	Guided Notes are PRIMARY · Science Journal is SECONDARY. Design notes first, then slides.
BQ/V/N mirror	Slides' Big Question / Vocabulary / Notes anchors mirror the student notes exactly.
Status vocabulary	AVAILABLE = exists · PLANNED = scheduled, not built (do NOT treat as a filename) · NOT REQUIRED · OPTIONAL/FLEX.
Friday FLEX	RECOVER (catch-up/absent) · PRACTICE (skill) · EXTEND (enrichment). Never blocks Monday.
Transfer vs transformation	Transfer = energy moves · transformation = energy changes form · BOTH is valid — never force either/or.
Same-object rule	Speed–energy comparisons always use the SAME object at different speeds.

READ FIRST Front matter (At a Glance → Pacing → Resource Map → Grading → Materials → Vocabulary → Demo Map) gives the whole unit. Chapter spreads give the day-by-day. Start with the Pacing Calendar.

HOW SLIDES + GUIDED NOTES WORK

RULE	WHAT IT MEANS
Notes-first	Guided Notes are designed BEFORE the matching slideshow. Notes are the primary student reference.
Exact mirror (BQ/V/N)	Every NOTES-mode slide matches the paper exactly – same section label, wording, bullets, blanks, order, diagram.
Scope of mirror	Mirror applies to NOTES-mode slides only. Not every blank needs a cue slide. Hooks/review MCQ are DEMO/DISCUSSION.
Reveal discipline	Fill-in answers are revealed, not front-loaded (prediction-before-reveal).
Vocabulary TEACH vs RECORD	TEACH = full-screen card (word + visual + exact source def + Reader page). RECORD = fill-in from memory, card covered.
9-step choreography	BIG QUESTION → PHENOMENON → VOCAB CARDS → VOCAB NOTES → READ → CHECKS → CONCEPT NOTES → APPLICATION → RETURN TO BQ
Mode badges	BQ · D (demo) · V (vocab) · R (read) · C (check) · N (notes) · W (write) · T (talk) · P (practice) · J (journal).
Notes navigation (permanent)	BQ / V1 / V2 / N1 / N2 / N3 / N4 – distinct from temporary HUD action bubbles.

STATUS Chapter 1 Guided Notes are FINAL (AVAILABLE). Chapters 2–7 notes and all slide decks are being rebuilt in Phases 3–4 (add per-section Reader page refs, chapter badge, full-screen vocab cards, canonical slide sequence). Chapter 1 spread references are real; Chapters 2–7 spread references remain PLANNED placeholders until then.

RESOURCE MAP

RESOURCE	AUDIENCE	REQUIRED	GRADED	STATUS	TEACHER KEY
Guided Notes (Ch1–7)	Student	CORE	No (formative)	Ch1 AVAILABLE/FINAL · Ch2–7 PLANNED	Ch1 AVAILABLE · Ch2–7 PLANNED
Science Journal prompts	Student	CORE	No	AVAILABLE	—
Chapter practice/application	Student	CORE	No (practice)	AVAILABLE	AVAILABLE
Chapter quiz (Ch1)	Student	CORE	Formative (graded)	Ch1 AVAILABLE (9-item)	AVAILABLE
Energy Detectives (Ch1)	Student	OPTIONAL/FLEX	No	AVAILABLE (reclassified)	AVAILABLE
Ramp It Up 2.0	Student (lab)	CORE	Formative	ADAPT — PLANNED	PLANNED
Pom-Pom Launcher	Student (build)	REQUIRED BUILD	Formative	PLANNED (net-new)	PLANNED
Study guide	Student	CORE	No (practice)	AVAILABLE	AVAILABLE
Review game	Student	CORE	No	AVAILABLE	AVAILABLE
Unit 1 test (25Q/50pt)	Student	CORE	GRADED	AVAILABLE (re-verify)	AVAILABLE
Teacher demos (7)	Teacher	CORE	—	AVAILABLE (content) / demo guide PLANNED	—
FLEX resources	Student/Teacher	OPTIONAL/FLEX	No	AVAILABLE	—
Absent recovery	Student/Teacher	CORE support	No	AVAILABLE	—
Media	Teacher	OPTIONAL/FLEX	—	PARTIAL (Ch1–2 staged · Ch3–7 scripts)	—

Full matrix with master locations: docs/playbook/UNIT1_RESOURCE_MAP.md

ASSESSMENT & GRADING PLAN

Unit 1 final (locked)

ATTRIBUTE	SPECIFICATION
Questions / points	25 questions · 50 points · 2 points each
Q1–8	Matching (8 terms, 11 definitions, 3 extras)
Q9–25	Multiple choice (4 options each)
Excluded	No short answer · no CER · no fill-in-the-blank
Key	Separate teacher key (never mixed into student pages)
Balance (by points)	Core 26% · data/diagram 30% · application 26% · engineering 18%

Routine scoring convention

> **SCORE** Score: _____ / [raw max] _____ % — All scored work is reported as raw score + percentage (out of 100).

Weighing

Teacher grading-weight policy: not specified by Unit materials. No assignment weights are invented here.

MASTER MATERIALS LIST

CATEGORY	ITEMS
Reusable classroom	Plastic ramp (or ruler + books) · toy car · plastic target cup · identical glass marbles · meter stick/ruler · shallow sand tray · thermometer · low-voltage battery circuit (LED/bulb/buzzer/motor) · flashlight · stopwatch/timer
Consumables	Index cards (vocab deck) · masking tape · craft kit (string, bells, cups, cardboard, rubber bands — latex-free) · warm water (heat demo) · sand
Chapter demos	Electric pencil sharpener + pencil (Ch1) · marbles + cup (Ch2) · ramp + marble + cup (Ch3) · cup/balloon/salt + flashlight + warm water + thermometer + AA/AAA circuit (Ch4) · toy car + cup (Ch5) · wind-up timer (Ch6) · flashlight (Ch7)
Ramp It Up 2.0	Standardized track (28×6 in, 1-in rails) · 3 identical books · same marble · lightweight target cup · printed 0–50 cm strip · ruler + tape · role cards (Launcher / Target Manager / Measurer / Recorder)
Pom-Pom Launcher	Craft kit: cardboard tube, rubber bands (latex-free option), pom-poms, tape · safety note: pieces to adult if band breaks
Optional / FLEX	Newton's cradle (teacher demo only) · solar cell (optional) · Energy Detectives scavenger hunt items · Drop It supplies

No specialty equipment invented beyond the locked demo/build requirements. Quantities scale by group count (class size is an open question).

UNIT VOCABULARY MAP (14 CORE VOCABULARY)

TERM	CH	READER P.	V-GROUP	NOTES STATUS	ASSESSMENT ELIGIBILITY	COMMON MISCONCEPTION
change	1	1	Ch1	Notes fill-in (Key Words)	Eligible (matching/MC)	–
energy	1	1	Ch1	Notes fill-in (Key Words)	Eligible (matching/MC)	"Energy is a visible substance"
stored energy	1	4	Ch1	Notes fill-in (Key Words)	Eligible (matching/MC)	"Stored means moving now" / "only moving objects have energy"
motion	2	7	Ch2	Notes fill-in	Eligible (matching/MC)	motion (process) ≠ energy of motion (possession)
energy of motion	2	9	Ch2	Notes fill-in	Eligible (matching/MC)	"faster always = more energy" (must be SAME object)
speed	3	13	Ch3	Notes fill-in	Eligible (matching/MC)	"ramp height = speed" (speed is inferred)
energy transfer	4	17	Ch4	Notes fill-in	Eligible (matching/MC)	"transfer = transformation" · "energy disappears"
collide	5	25	Ch5	Notes fill-in	Eligible (matching/MC)	"energy disappears after collision"
collision	5	25	Ch5	Notes fill-in	Eligible (matching/MC)	same as collide
sound	5	28	Ch5 (L10)	Notes fill-in	Eligible	"sound is

TERM	CH	READER P.	V-GROUP	NOTES STATUS	ASSESSMENT ELIGIBILITY	COMMON MISCONCEPTION
					(matching/MC)	matter moving"
convert	6	32	Ch6	Notes fill-in	Eligible (matching/MC)	transfer vs transformation
conversion	6	32	Ch6	Notes fill-in	Eligible (matching/MC)	transfer vs transformation
energy transformation	6	32	Ch6	Notes fill-in	Eligible (matching/MC)	transfer vs transformation
engineering design	6	36	Ch6	Notes fill-in	Eligible (matching/MC)	"biggest/ loudest = best"

NOT REQUIRED Kinetic · potential · gravitational potential · mechanical · radiant · momentum · inertia · compression wave are NOT introduced as required (TG explicitly avoids them).

DEMO / ACTIVITY / LAB / BUILD MAP

CHAPTER	EXPERIENCE	CATEGORY
Ch1	Electric Pencil Sharpener	DEMO
Ch2	Marble Collision	DEMO
Ch3	Ramp + Target Preview	DEMO
Ch3	Ramp It Up 2.0 (Day 1 + Day 2)	LAB / INVESTIGATION
Ch4	Balloon Drum / salt (sound transfer)	DEMO
Ch4	Source → Path → Receiver → Evidence	ACTIVITY
Ch5	Toy-Car Collision	DEMO
Ch5	Collision application / before-after models	ACTIVITY
Ch6	Wind-Up Timer	DEMO
Ch6	Transfer / Transformation / Both sort	ACTIVITY
Ch7	Flashlight	DEMO
End of unit	Pom-Pom Launcher	REQUIRED BUILD
Post-assessment	Open-ended engineering challenge	POST-TEST STEM
–	Energy Detectives · Moving Marbles · Drop It	OPTIONAL / FLEX

CHAPTER TEACHER SPREAD — TEMPLATE (BLANK)

Reusable two-page spread. LEFT = Teach It · RIGHT = Teacher Support. Populate in place; do not overfill.

LEFT PAGE — TEACH IT

Chapter number / title: _____

Big Question: _____

Reader pages: _____

Approximate instructional time: _____

Student resources: _____

Vocabulary / V-groups: _____

N-anchor map: _____

Lesson blocks / sequence: _____

What to project: _____

What students do: _____

RIGHT PAGE — TEACHER SUPPORT

Demo / activity:

Materials:

Setup:

Teacher moves:

Ask:

Listen For:

Misconception / corrective move:

Do-not-overteach:

If short on time:

Collect / grade:

Absent / makeup:

What comes next:

CHAPTER 1 — TEACHER SPREAD (PROTOTYPE)

CHAPTER 1 · ENERGY CAUSES CHANGE

LEFT PAGE — TEACH IT

FIELD	CONTENT
Big Question	Where can we observe evidence of energy causing change? (TG Lesson 2 adds "What are some forms of energy?" → addressed by N2 on the notes)
Reader pages	Printed 1–6 · PDF 7–12 (core instruction pp. 1–5; p. 6 = forward preview "Energy Can Move and Change")
Approximate time	Two 30-min blocks (Block 1 "Observe & Name" · Block 2 "Forms & Stored")
Student resources	Guided Notes — AVAILABLE/FINAL (01-ch1-energy-causes-change-guided-notes, 3 pp) · practice — AVAILABLE · Energy Detectives (FLEX) — AVAILABLE
Vocabulary / V-groups	V1: energy · change (Reader p. 1) · V2: stored energy (Reader p. 4). evidence + position = Language of Instruction (no Reader page) → concept blanks in N1/N3.
N-anchor map	N1 Energy Causes Change (ENERGY→CHANGE→EVIDENCE) · N2 Energy Occurs in Different Forms (5-form table) · N3 Objects Can Store Energy (STORED→RELEASE→CHANGE + examples + position) · N4 Recognizing Evidence (BEFORE/AFTER + CHECK YOUR THINKING) · BQ · NOW I THINK
Lesson blocks (Block 1)	BQ → sharpener DEMO → discuss change → vocab cards (energy, change) → read pp. 1–2 → MODEL IT + DRAW/LABEL → exit
Lesson blocks (Block 2)	recall → read p. 3 (forms) → vocab card (stored) → read pp. 4–5 → FORMS + STORED + evidence table → application → return to BQ
What to project	01-ch1-energy-causes-change-slides.pptx — FINAL (45 slides, 16:9) · BQ 2/3/33 · V1 10–12 · V2 20–22 · N1 13–16 · N2 17–19 · N3 23–27 · N4 28–31 · demo 4, 6–9 · review 34–

	45 · Reader pp. 1–6
What students do	Predict → observe → fill Key Words → complete ENERGY→CHANGE→EVIDENCE model → draw before/after → record evidence → answer Big Question

RIGHT PAGE — TEACHER SUPPORT

FIELD	CONTENT
Demo / activity	Electric Pencil Sharpener (live DEMO). Flashlight + cup/balloon/salt as secondary demos.
Materials	Electric sharpener + unsharpened pencil · flashlight + D-cell · cup + balloon + salt
Setup	Sharpener on stable surface · balloon drumhead pre-stretched · flashlight checked
Teacher moves	Ask "What changed?" before naming energy · record every change · build model on the board
Ask	"What changed?" "What makes it happen?" "How is the change evidence of energy?"
Listen For	"pencil got shorter/sharper" "it made noise" — evidence, not "I saw the energy"
Misconception / fix	"Only moving objects have energy" → pair moving + stored examples. "Energy is a visible substance" → we see the change, not the energy.
Do-not-overteach	Do not introduce transfer/transformation or kinetic/potential. Keep p. 6 as a forward preview only.
If short on time	Cut the cup/balloon/salt demo · move Energy Detectives to Friday FLEX.
Collect / grade	Exit ticket (formative) · notes completion (not graded). No scored chapter item.
Absent / makeup	Absent-student recovery guide — curriculum/chapter-01/student/recovery/01-ch1-absent-student-recovery.md
What comes next	Chapter 2 — Moving Objects Have Energy (marble collision demo).

CHAPTER 1 DEMO GUIDE – ELECTRIC PENCIL SHARPENER

LIVE Live demo is preferred. Virtual backup is OPTIONAL / DEFERRED (the previously attempted AI-generated sharpener video failed and must NOT be reused as a dependency).

SECTION	GUIDANCE
Materials	Electric pencil sharpener · unsharpened pencil · optional second (dull) pencil for BEFORE/AFTER side-by-side
Set Up	Sharpener on a stable, visible surface. Have one dull and one sharp pencil ready to compare.
Do This	1) Show the dull pencil. 2) Ask students to predict what will change. 3) Sharpen the pencil. 4) Have students name every change (shorter, sharper, noise).
Ask	"What changed?" "What did you notice with your eyes? with your ears?" "What made the change happen?"
Listen For	"The pencil got shorter and sharper." "The sharpener made noise." Evidence language, not "I saw energy."
Connect	Each change is evidence of energy. Energy = the ability to cause change (Reader p. 1). Build ENERGY → CHANGE → EVIDENCE on the board.
Do Not Overteach	Do not claim the tip warms up as student-observed heat (TG "Know the Science 1" – optional, teacher-observed only). No "mechanical/electrical energy" required terms.
If Students Ask	"Where does the energy come from?" → "It comes through the cord/battery – we'll learn more about that later." Keep it a preview.
If Short on Time	Skip the side-by-side comparison · use a single sharpen. Fall back to "flip the light switch" as a backup change.

CHAPTER 1 — SLIDESHOW MAP (FINAL)

FILE production/slides/01-ch1-energy-causes-change-slides.pptx — 45 slides (16:9). Rendered PDF + 45 PNG previews alongside. Speaker notes on every slide (MODE · TIME · ASK · LISTEN FOR · REVEAL · NEXT).

ANCHOR	SLIDES	WHAT HAPPENS ON SCREEN
BQ	2 · 3 · 33	2 = Big Question banner + TURN & TALK · 3 = My first thinking (write) · 33 = BQ · NOW I THINK (return, write)
V1 (energy · change)	10–12	10 = two vocab cards · 11 = notes write · 12 = notes reveal (change · different)
V2 (stored energy)	20–22	20 = one large vocab card · 21 = notes write · 22 = notes reveal (later)
N1 Energy Causes Change	13–16	13 = definition · 14 = ENERGY → CHANGE → EVIDENCE model · 15 = write · 16 = reveal (cause · change · evidence)
N2 Energy Occurs in Different Forms	17–19	17 = five forms · 18 = clap check · 19 = 5-row form/example table
N3 Objects Can Store Energy	23–27	23 = still-battery predict · 24 = position · 25 = STORED → RELEASE → CHANGE model · 26 = write · 27 = reveal (position)
N4 Recognizing Evidence	28–31	28 = evidence concept · 29 = BEFORE/AFTER draw · 30 = CHECK YOUR THINKING write · 31 = sample + classify
Phenomenon / demo	4 · 6–9	4 = light-switch change · 6 = observe dull pencil · 7 = predict · 8 = before/after · 9 = change = evidence
Application / evidence	32	Which change shows energy at work? (choose the change)
End-of-chapter review	34–45	6 multiple-choice (A/B/C/D) — question

ANCHOR	SLIDES	WHAT HAPPENS ON SCREEN
		state then answer state; letters shuffled A/B/C/D/B/C

Reader cues on-slide: V1/N1 pp. 1–2 · N2 p. 3 · V2/N3 pp. 4–5 · N4 pp. 1–6 · BQ p. 1. Timing: two 30-min blocks (Block 1 = slides 1–12 · Block 2 = slides 13–33) + review slides 34–45 as a third short block or Friday FLEX.

CHAPTER 1 — SUPPORTING RESOURCE INDEX (PHASE 5)

> **PRINCIPLE** Lean ecosystem: every resource has a distinct instructional job; nothing merely repeats the Guided Notes. Guided Notes (primary) + Science Journal (secondary) + a separate worksheet only for a distinct apply task.

RESOURCE	CANONICAL FILE	REQUIRED	GRADED	WHEN USED	KEY / ABSENT
Guided Notes (3 pp)	01-ch1-energy-causes-change-guided-notes	REQUIRED	Not graded	Blocks 1–2 (notes write)	key AVAILABLE
Slideshow (45 slides)	01-ch1-energy-causes-change-slides.pptx	REQUIRED	—	Blocks 1–2 + review	— (CHECK slides reveal)
Practice / application	01-ch1-energy-causes-change-practice.docx	REQUIRED	Not graded (practice)	Block 2 application	key AVAILABLE
Chapter quiz (9-item)	01-ch1-energy-causes-change-quiz.docx	REQUIRED	Formative (graded)	End of chapter	key + blueprint AVAILABLE
Science Journal	MASTER_CURRICULUM_MAP.md (2 prompts)	OPTIONAL/FLEX	Not graded	Daily, secondary role	— (reflective)
Absent / makeup	student/recovery/01-ch1-absent-student-recovery.md	REQUIRED (support)	Not graded	After absence	self-contained
Energy Detectives	01-ch1-energy-detectives-student.docx	OPTIONAL/FLEX	Not graded	Friday FLEX	key AVAILABLE
Friday FLEX	student/01-ch1-friday-flex.md	OPTIONAL/FLEX	Not graded	Friday	—
Study-guide contribution	docs/chapter-01/CH1_STUDY_GUIDE_CONTRIBUTIO	REFERENCE	—	Unit assembly	—

RESOURCE	CANONICAL FILE	REQUIRED	GRADED	WHEN USED	KEY / ABSENT
	N.md				
Assessment trace	docs/chapter-01/CH1_ASSESSMENT_TRACE.md	REFERENCE	–	Unit assembly	–

LEAN No new student handout or teacher key was created in Phase 5. The practice page is a distinct APPLY task (evidence judgment, not note recopy), so it is kept. Science Journal is already covered by two daily prompts. Energy Detectives is reclassified OPTIONAL/FLEX.

ANSWER-KEY ARCHITECTURE

How keys are built

PRINCIPLE	REQUIREMENT
Visual mirror	Keys mirror student pages exactly — same layout, page numbers, section codes (BQ / V1 / V2 / N1 / N2).
Fill directly	Answers written onto the mirrored page; no separate answer list.
Accepted/sample responses	Open items: sample response + range of acceptable alternatives (TEACHER_KEY_STANDARD).
Scoring notes	Only when needed: partial credit, point allocation, what NOT to penalize.
Separation	Keys are never mixed into student PDFs; separate teacher file per assessment/quiz/worksheet/lab.

Key index / status

KEY	RESOURCE	STATUS
Ch1 practice key	01-ch1-energy-causes-change-practice-key	AVAILABLE
Ch1 quiz key	01-ch1-energy-causes-change-quiz-key	AVAILABLE
Ch1 guided-notes key	01-ch1-energy-causes-change-guided-notes-key	AVAILABLE (FINAL · 3 pp)
Ch1 Energy Detectives key	01-ch1-energy-detectives-key	AVAILABLE (OPTIONAL/FLEX)
Ramp It Up 2.0 key	03-ch3-ramp-it-up (ADAPT)	PLANNED / NOT YET GENERATED (update existing)

KEY	RESOURCE	STATUS
Pom-Pom Launcher key	net-new build	PLANNED / NOT YET GENERATED
Study guide key	unit1 study guide	AVAILABLE
Review game guide	unit1 review game	AVAILABLE
Unit 1 final key	07-unit-final-assessment-key	AVAILABLE (re-verify after note rebuild)

SPECIAL SECTIONS — STATUS (PHASE 2)

SECTION	STATUS	POINTER
Ramp It Up 2.0 teacher guide	SKELETON · ADAPT — PLANNED	production/labs/03-ch3-ramp-it-up-teacher-spec.md
Pom-Pom Launcher teacher guide	SKELETON · PLANNED (net-new)	net-new (ADAPTED from colleague)
Review plan	SKELETON · points to AVAILABLE assets	production/review/unit-01/ (study guide + game)
Unit 1 assessment administration	SKELETON · locked 25Q/50pt	production/quizzes/07-unit-final-assessment.docx
Answer-key index	RENDERED	see Answer-Key Architecture
FLEX / enrichment / catch-up	SKELETON · points to framework	docs/planning/ FRIDAY_FLEX_FRAMEWORK.md
Absent / makeup support	Ch1 AVAILABLE · Ch2–7 SKELETON	curriculum/chapter-0N/student/ recovery/
Media index	SKELETON · Ch1–2 staged · Ch3–7 scripts-only	docs/planning/ WHOLE_UNIT_MEDIA_STAGING.md

CHAPTERS 2–7 — SPREAD PLACEHOLDERS

Each chapter will use the reusable spread template (LEFT Teach It / RIGHT Teacher Support). Populated in later phases.

CHAPTER 2 · MOVING OBJECTS HAVE ENERGY

BIG QUESTION	READER PP.	DEMO	NEW CORE VOCABULARY
How are energy, change, and movement of objects related?	7–12	Marble Collision	motion, energy of motion

CHAPTER 3 · ENERGY AND SPEED ARE RELATED

BIG QUESTION	READER PP.	DEMO	NEW CORE VOCABULARY
How are energy and speed related?	13–16	Ramp + Target Preview	speed

CHAPTER 4 · ENERGY TRANSFER

BIG QUESTION	READER PP.	DEMO	NEW CORE VOCABULARY
What evidence shows that energy transfers from place to place?	17–24	Balloon Drum / salt (sound)	energy transfer

CHAPTER 5 · COLLISIONS TRANSFER ENERGY

BIG QUESTION	READER PP.	DEMO	NEW CORE VOCABULARY
How is energy involved in collisions? · What happens when objects collide? How are collisions predictable?	25–30	Toy-Car Collision	collide, collision, sound

CHAPTER 6 · HOW CAN ENERGY TRANSFORMATIONS SOLVE PROBLEMS?

BIG QUESTION	READER PP.	DEMO	NEW CORE VOCABULARY
How do energy transformations help people? · How can I design a device that transforms energy to solve a problem?	31–36	Wind-Up Timer	convert, conversion, energy transformation, engineering design

CHAPTER 7 · SOLVING PROBLEMS AND DESIGNING SOLUTIONS: THOMAS A. EDISON

BIG QUESTION	READER PP.	DEMO	NEW CORE VOCABULARY
How did Thomas Edison use his knowledge of energy transfer and transformation to solve problems? · What have I learned about energy?	37–42	Flashlight	(synthesis — no new Core Vocabulary)